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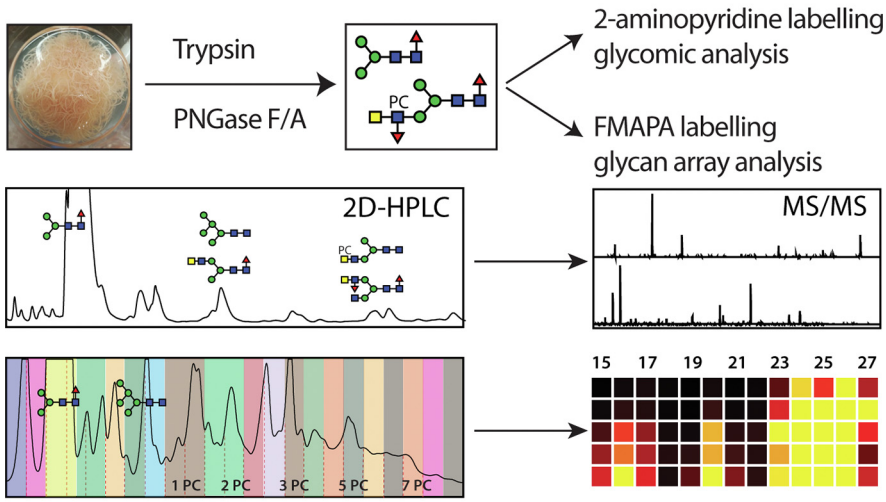
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In Brief

Glycan-dependent host immunomodulation has been previously reported in the case of the porcine parasite *Trichuris suis*. Using off-line HPLC/MALDI-TOF-MS/MS, we reveal a high degree of N-glycomic variety with LacdiNAc-type antennae modified with fucose and/or phosphorylcholine residues as well as HexNAc-substituted fucose. A natural glycan array was also generated whose application showed that selected fractions of HPLC-separated N-glycans were recognized by mammalian innate immune lectins and antibodies in the sera of infected pigs.

Graphical Abstract



Highlights

- The porcine whipworm, *Trichuris suis*, has a complex and unique N-glycome.
- MS/MS reveals many LacdiNAc motifs substituted with fucose and phosphorylcholine.
- Glycan array data show recognition by immune lectins and infection sera.

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Recognition of Highly Branched N-Glycans of the Porcine Whipworm by the Immune System

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Glycans are key to host–pathogen interactions, whereby recognition by the host and immunomodulation by the pathogen can be mediated by carbohydrate binding proteins, such as lectins of the innate immune system, and their glycoconjugate ligands. Previous studies have shown that excretory-secretory products of the porcine nematode parasite *Trichuris suis* exert immunomodulatory effects in a glycan-dependent manner. To better understand the mechanisms of these interactions, we prepared N-glycans from *T. suis* and both analyzed their structures and used them to generate a natural glycan microarray. With this array, we explored the interactions of glycans with C-type lectins, C-reactive protein, and sera from *T. suis*–infected pigs. Glycans containing LacdiNAc and phosphorylcholine-modified glycans were associated with the highest binding by most of these proteins. In-depth analysis revealed not only fucosylated LacdiNAc motifs with and without phosphorylcholine moieties but phosphorylcholine-modified mannose and *N*-acetylhexosamine–substituted fucose residues, in the context of maximally tetraantennary N-glycan scaffolds. Furthermore, O-glycans also contained fucosylated motifs. In summary, the glycans of *T. suis* are recognized by both the innate and adaptive immune systems and also exhibit species-specific features distinguishing its glycome from those of other nematodes.

Nematodes or roundworms represent a large group of invertebrates whose species-specific habitats range from the soil through to the mammalian gut. A large number of nematodes are parasitic and their ability to often survive years despite recognition by host immune systems is remarkable but probably reflects co-evolution of the immunomodulatory capacity of the nematode and the response of the host. The ability of parasites to shift the balance of the host immune system is due to a range of molecules, some of which are glycoproteins (1). The dichotomy between recognition as 'foreign' and a lack of full expulsion or eradication suggests that an expected immune response is misdirected or aborted. One example of this is the role of phosphorylcholine on nematode glycoconjugates; this zwitterionic moiety is well known as a component of phosphatidylcholine, a typical class of eukaryotic phospholipids and of platelet-activating factor, a related signaling molecule (2) but is also a widespread modification of glycoconjugates in nonvertebrate species, including lipopolysaccharides, N-glycans, and glycolipids (3, 4). In the case of bacterial infections in mammals, recognition of phosphorylcholine-modified glycoconjugates *via* C-reactive protein or antiphosphorylcholine antibodies is associated with complement-mediated killing of the pathogen (5). In contrast, N-glycans carrying phosphorylcholine, exemplified by those on the ES-62 excretory-secretory protein of the rodent parasite *Acanthocheilonema viteae*, may be conformationally more flexible than bacterial polysaccharides; thus, despite binding of an ES-62/C-reactive protein complex to C1q, the full complement cascade response is not activated (6). On the other hand, ES-62 is just one example of a nematode protein with immunomodulatory effects (7); indeed many helminth parasites alter the balance between various T cell responses, so aiding parasite survival but also affecting other aspects of the immune system (8).

Other than phosphorylcholine, N-glycans from nematodes are often fucosylated either on the core region or the antennae. One antennal motif, the so-called LDNF epitope (fucosylated LacdiNAc; GalNAcβ1,4[Fucα1,3]GlcNAc), is present in some nematodes (9) as well as schistosomes, which are trematode parasites (10); this motif is immunogenic in mammals (11, 12) but is a mimic of the Lewis X epitope present on a range of mammalian glycans, including sialylated and sulfated forms recognized by selectins with roles in the immune system. In nematodes and lepidopteran insects (13), LDNF on N-glycans can also occur in a phosphorylcholine-modified form, as exemplified by structures we previously found in *Trichuris suis* (14), a porcine parasite having a negative effect on pig productivity and

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relative of the human parasite *Trichuris trichiura*, with an estimated 1 billion infected individuals (15). Further modifications of nematode glycans include glucuronylation of the antennae, fucosylation of the second (distal) core *N*-acetylglucosamine residue, and galactosylation of core fucose residues (9, 16), while sialylation, sulphation, and phosphorylation are not known features of nematode N-glycans. However, all these variations do not exist in a single organism and the glycan modifications can be differently combined resulting in a high degree of interspecies N-glycomic variability. In comparison to nematode N-glycomes, there is even less published information on their O-glycomes, other than those of *Caenorhabditis elegans* (17–19) or *Toxocara canis* (20), whereby the methylated O-glycans from the latter are an epitope for antibodies of infected animals (21).

Considering the observed immunomodulatory effects of nematode glycoconjugates, links have been made between reduced nematode infections and increased occurrence of allergies and autoimmunity in humans living within more developed societies (22). As a parasite that generally does not have a productive life-cycle in humans, *T. suis* has attracted attention as a potential unorthodox therapy for such diseases and its eggs have been administered to patients with Crohn's disease and autoimmune rhinitis as well as to animals with experimental autoimmune encephalomyelitis (23–25). While the effects and safety of such therapies are controversial (26), it is clear that *T. suis* excretory-secretory products do exert typical immunomodulatory effects, which are partly glycan-dependent (27). Examples of proteins secreted by *Trichuris* spp. include enzymes or the cytokine-binding protein p43, which has also been explored as a vaccine candidate (28). With this background and to investigate interactions of *T. suis* glycans with components of mammalian immune systems, we have established the first microarray of natural glycans of this species, accompanied by an in-depth study of its N-glycomic capacity. Thereby, we show that especially the larger N-glycans bind mammalian lectins as well as antibodies present in the sera of infected pigs. Additionally, the protein corresponding to the major lectin-binding band, observed by Western blotting, as well as the O-glycans were analyzed.

EXPERIMENTAL PROCEDURES

Biological Material

Adult *T. suis* worms were kindly provided by Dr Irma van Die (Amsterdam University Medical Center). The parasites were isolated from pigs 3 months post infection; one part of the harvested parasites was maintained in media before further treatment. In total, up to 29 g (wet weight) of worms were used for preparation. *T. suis* pooled uninfected and infected pig sera from different time points of infection (21 days, 28 days, 35 days, and 49 days post infections) for glycan array analysis were kindly provided by Dr Andrew Williams (University of Copenhagen).

Enzymatic Release and N-Glycan Purification for Glycomic and Array Analyses

The worms were thoroughly washed with 0.98% (w/v) NaCl solution using a 0.22 μm filter device (Millipore) prior to lyophilization and grinding in liquid nitrogen. The dry powder was dissolved in 20 ml lysis buffer (25 mM Tris, 150 mM NaCl, 5 mM EDTA, and 0.5% (w/v) CHAPS, pH 7.4) followed by multiple rounds of sonication. The insoluble fraction was removed by centrifugation and the supernatant was dialyzed prior to lyophilization. The dried glycoproteins were reduced using 1,4-DTT (Sigma-Aldrich; 0.18 M) followed by carboxymethylation using iodoacetamide (Sigma-Aldrich; 0.18 M). The reduced and carboxymethylated *T. suis* protein homogenates were proteolyzed using trypsin (Sigma-Aldrich, Trypsin Sequencing Grade from bovine pancreas, pH 8.4; 1 mg/ml), prior to solid-phase extraction (SPE) on 2 g Sep-Pak C18 (Waters). The glycopeptides were eluted from the C18 material with increasing 1-propanol concentration (20%, 40%, and finally 100%). The eluates were combined and lyophilized overnight. Thereafter, the N-glycans were released using peptide:N-glycosidase F (New England Biolabs, in 50 mM ammonium bicarbonate, pH 8.4; twice 250 U/ml overnight) followed by peptide:N-glycosidase A (New England Biolabs, in 50 mM ammonium acetate, pH 4.0; 60 U/ml). The free glycans were separated from residual O-glycopeptides by SPE on C18. The glycans were eluted using 5% acetic acid and the residual peptides with increasing 1-propanol concentration (20%, 40%, and finally 100%). The free N-glycans were analyzed by MALDI-TOF MS (UltraFlex II MALDI/TOF Mass Spectrometer, Bruker Daltonics, equipped with a Smartbeam II laser) and were either permethylated (as an initial screen), labeled with FMAPA prior to glycan array preparation, or labeled with 2-aminopyridine for in-depth off-line LC-MS glycomic analyses as described below.

FMAPA-Labeling of N-Glycans Prior to Glycan Array Preparation

Ninety percent of the free N-glycans were conjugated to the fluorescent Fmoc-(3-(methoxyamino)propylamine) linker (abbreviated as FMAPA) as described by Wei *et al.* (29). Briefly, the free N-glycans were incubated with 0.35 M Fmoc-MAPA linker, 0.5 M sodium acetate, and 0.1 M 2-amino-5-methoxybenzoic acid (2-AM) at 65 °C for 4 h. Afterward, a ten-fold excess of ethylacetate was added and incubated at −20 °C for at least 20 min to precipitate the neoglycoconjugates. The pellet was washed 3 times prior to SPE on C18. The successful labeling of the *T. suis* N-glycans was determined by MALDI-TOF MS. The Fmoc-MAPA–labeled N-glycans were fractionated by a semipreparative normal phase HPLC (Luna, Phenomenex, LC Column, 250 × 10 mm, 5 μm, 100 Å) on a Shimadzu HPLC CBM-20A system, equipped with a UV detector (SPD-20AV) and a fluorescence detector (RF-20A). The following gradient was applied using a three-buffer system (buffer A – 250 mM ammonium acetate, pH 4; buffer B – dH_2O HPLC grade water; buffer C – 99% (v/v) acetonitrile) at a flow rate of 4 ml/min: 0 to 5 min, 16% B and 4% C; 5 to 50 min, 16 to 40% B and 4 to 50% C; 60 to 60.1 min back to starting conditions and hold for another 10 min. Selected peaks were re-injected on a second dimension using a reversed phase analytical C18 – HPLC column (Agilent Zorbax Eclipse Plus 95 Å C18, 150 × 2.1 mm, 5 μm, analytical). The following gradient of buffer D (water) and buffer E (0.1% (v/v) TFA in acetonitrile) was applied at a flow rate of 1 ml/min: 0 to 4 min, 5 to 50% E; 40 to 45 min, hold 50% E; 45 to 46 min, 50 to 60% E; 46 to 51 min, hold 60% E; 51 to 52 min, back to starting conditions and hold for another 8 min. The detector was set to Ex/Em 265/315 nm. The column was calibrated daily in terms of in-house–prepared standard Lacto-*N*-neotetraose FMAPA glycans. The Fmoc moieties were released by incubating the labeled N-glycans with 5% piperidine in water for 30 min at room temperature under constant shaking. The

reagent was removed by chloroform precipitation prior to C18-SPE purification before printing on NHS-activated slides.

N-Glycan Array Preparation and Probing

The MAPA-glycans were diluted to 100 μM in spotting buffer (100 mM sodium phosphate, pH 8.5) and printed using a sciFLEXARRAYER S11 (Scienion) on NHS-activated glass slides (Nexterion, Slide H, Schott) by noncontact piezo printing. For each sample, four spots were printed per array. After 16 h of hybridization, slides were blocked (50 mM ethanolamine in 100 mM sodium borate, pH 9.0) and subsequently rinsed with TSM-WB (1xTSM, 0.05% Tween-20) and water. The slides were sealed and stored dry at –20 °C until use. The slides were rehydrated in TSM buffer (20 mM Tris–Cl, 150 mM sodium chloride, 2 mM calcium chloride, and 2 mM magnesium chloride) before incubation with (i) 10 μg/ml biotinylated lectins from either *Canavalia ensiformis* (Concanavalin A; ConA), *Aleuria aurantia* (AAL), *Galanthus nivalis* (GNA), *Lotus tetragonolobus* (LTL), *Wisteria floribunda* (WFA), *Triticum aestivum (wheat germ agglutinin; WGA)*, *Ricinus communis* (RCA), or *Sambucus nigra* (SNA) from VectorLabs (in 1× TSM, 0.05% Tween-20, and 1% BSA, *i.e.*, TSMBB) followed by 2 μg/ml anti-biotin antibody AF 488 conjugated (Invitrogen) in TSMBB; (ii) 2 μg/ml anti-LDNF (L6B8 in TSMBB) (30) and anti-FLDNF (F2D2 in TSMBB) (31) followed by anti-mouse IgG AF 647 conjugated (1:1000, Invitrogen) in TSMBB or 5 μg/ml anti-mannose antibody (100-4G11-A) (32) followed by 2 μg/ml anti-mouse IgM AF 647 (Invitrogen); (iii) 5 μg/ml human C-reactive protein (CRP; biotechne) in TSMBB with additional 2 mM Ca^{2+} followed serially by anti-CRP from mouse (biotechne) in TSMBB and 2 μg/ml anti-mouse IgG AF 647 in TSMBB; (iv) 10 μg/ml TEPC-15 (Sigma-Aldrich) followed by 2 μg/ml anti-mouse IgA FITC in TSMBB; (v) 5 μg/ml human DC-SIGN-Fc (R&D Systems), 5 μg/ml human MGL-Fc (R&D Systems), and 1 μg/ml human Dectin-2-Fc (Sino Biological) in TSMBB followed by 5 μg/ml anti-human IgG AF 488 (Invitrogen); (vi) 1:250 dilution of uninfected and infected pig sera (provided by Dr Andrew Williams, University of Copenhagen) followed by mouse anti-pig IgG or IgM antibody (1:500 diluted in TSMBB). Binding was detected by 2 μg/ml anti-mouse IgG AF 647 (Invitrogen). After each incubation step, slides were serially washed with TSM-WB, TSM and water. The dried slides were scanned with a GenePix 4300A Scanner (Molecular Devices) equipped with multiples lasers with a laser power from 10 to 100% and PMT gain from 450 to 600, and the images processed using the GenePix Pro 7.2 software (Molecular Devices). The raw fluorescent values were automatically normalized and the background was automatically subtracted from each individual spot. The generated data files (gpr file) were further analyzed by Excel (Microsoft). The average relative fluorescence units were calculated as averaged of all fluorescence values of each probe (*i.e.*, 4 spots) and the standard deviations were calculated and plotted. Probes with generally less than 500 relative fluorescence units in the assays and/or poor MS data were excluded from the presented data; heat maps were generated using the GLAD tool (33).

In-Depth Analysis of Pyridylaminated N-Glycans

The remaining 10% of the free N-glycans (see above) was subject to solid-phase extraction on non-porous graphitized carbon (Supel-Clean ENVICarb, Sigma-Aldrich) and eluted with 40% acetonitrile. N-glycans were labeled *via* reductive amination using 2-aminopyridine (PA). For 1D-HPLC, 10% of the pyridylaminated N-glycome was fractionated by HIAX-HPLC (IonPac AS11 column, Dionex, 4 × 250 mm, combined with a 4 × 50 mm guard column) using a two-solvent gradient (buffer A, 0.8 M ammonium acetate (pH 3.85) and buffer B, 80% acetonitrile, LC-MS grade) at a flow rate of 1 ml/min as follows: 0 to 5 min, 99% B; 5 to 50 min, 90% B; 50 to 65 min, 80% B; 65 to 85 min, 75% B. The HIAX-HPLC was calibrated using oligomannosidic PA-labeled bean glycans; detection was by fluorescence at 320/400 nm (excitation/emission). All manually collected HPLC glycan fractions were analyzed after lyophilization by MALDI-TOF MS and MS/MS. Selected fractions were combined and subject to a 2D-HPLC analysis by reversed-phase HPLC (Ascentis Express RP-amide, Sigma-Aldrich; 150 × 4.6 mm, 2.7 μm) and a gradient of 30% (v/v) methanol (buffer B) in 100 mM ammonium acetate, pH 4 (buffer A), was applied at a flow rate of 0.8 ml/min (Shimadzu LC-30 AD pumps) as follows: 0 to 4 min, 0% B; 4 to 14 min, 0 to 5% B; 14 to 24 min, 5 to 15% B; 24 to 34 min, 15 to 35% B; 34 to 35 min, return to starting conditions. The RP-amide HPLC column was calibrated daily in terms of glucose units using a pyridylaminated dextran hydrolysate and the degree of polymerization of single standards was verified by MALDI-TOF MS. Monoisotopic MALDI-TOF MS was performed using an Autoflex Speed (Bruker Daltonics) instrument in either positive or negative reflection mode with 6-aza-2-thiothymine or 2,5-dihydroxybenzoic acid as matrix. In general, MS/MS was performed by laser-induced dissociation of $[M + H]^+$ or $[M-H]^-$ ions; typically 2000 shots were summed for MS (reflector voltage, lens voltage, and gain of 27 kV, 9 kV, and 2059 V, respectively) and 4000 for MS/MS (reflector voltage, lift voltage, and gain of 27 kV, 19 kV, and 2246 V, respectively). Spectra were processed with the manufacturer's software (Bruker Flexanalysis 3.3.80) using the SNAP algorithm with a signal/noise threshold of 6 for MS (unsmoothed) and 3 for MS/MS (smoothed four times). Glycan spectra were manually interpreted on the basis of the masses of the predicted component monosaccharides, differences of mass in glycan series, comparison with coeluting structures from other insects, marine organisms or other nematodes, and fragmentation patterns before and after chemical treatment or exoglycosidase digestion. Assigned glycans had an interpretable MS/MS spectrum with at least three fragment ions, including either a Y1-ion at *m/z* 300 or 446 corresponding to a pyridylaminated reducing core $GlcNAc_1Fuc_{0-1}$ or (in the case of phosphorylcholine-modified glycans) the neutral loss of 299 or 445 Da (*i.e.*, loss of reducing terminal $GlcNAc_1Fuc_{0-1}PA$). A list of theoretical *m/z* values for each glycan composition is presented in supplemental Tables S1 and S2.

Enzymatic and Chemical Treatments of Pyridylaminated N-Glycans

Glycans were treated, prior to re-analysis by MALDI-TOF–MS, with α-fucosidase (bovine kidney from Sigma-Aldrich or almond α1,3/4-specific from Prozyme), α-mannosidase (jack bean from Sigma-Aldrich), or β-*N*-acetylhexosaminidases (jack bean from Sigma-Aldrich, *Streptomyces plicatus* chitinase from New England Biolabs or in-house–produced recombinant *C. elegans* HEX-4 specific for β1,4-GalNAc-linked residues) in 50 mM ammonium acetate, pH 5, at 37 °C overnight (except for pH 6.5 in the case of HEX-4). Hydrofluoric acid was used for removal of core or antennal α1,3-fucose, phosphorylcholine or phosphate. As appropriate, treated glycans were re-chromatographed by RP-amide HPLC to ascertain retention time shifts prior to MALDI-TOF-MS.

O-Linked Glycan Analysis by β-Elimination

Fifty-five milligrams of sodium borohydride ($NaBH_4$, Sigma-Aldrich) were dissolved in 1 ml 0.1 M NaOH solution. Four hundred microliters were added to a fraction of the completely dried residual *T. suis* glycopeptides after PNGase F/A release and incubated over night at 45 °C. The reaction was stopped by adding pure acetic acid. The mixture was initially purified by a cation exchange material (Dowex AG50 H^+ form, Bio-Rad). The free O-glycans were released by 5% acetic acid, lyophilized prior to co-evaporation using 10% acetic acid in methanol. This step was repeated at least twice to remove remaining salts prior to further purification using SPE on C18. Briefly, the sample was resuspended in 50% methanol and loaded on the column. The O-

glycans were eluted with 5% acetic acid and further permethylated prior to analysis by MALDI-TOF MS.

Glycan Permethylation

For permethylation, *T. suis* N- and O-glycans were dried in a glass tube and resuspended in 1 ml of a slurry of grinded NaOH pellets (Sigma-Aldrich) in DMSO followed by adding 500 µl iodomethane. The mixture was incubated for 20 min at room temperature under constant shaking and then the reaction was quenched using 200 µl dH_2O. Subsequently, permethylated N- and O-glycans were extracted in chloroform by constant washing with dH_2O and then applied to a pre-equilibrated C18 SPE column. Salts and contaminants were removed using 15% (v/v) acetonitrile prior to elution of the permethylated glycans using a 50% (v/v) acetonitrile solution. MALDI-TOF MS was performed using an Autoflex Speed or a Rapiflex (Bruker Daltonics) instrument in positive reflection mode with 2,5-dihydroxybenzoic acid as matrix.

Western Blotting

Fifteen micrograms of worm proteins were denatured at 95 °C in 5 × SDS loading buffer (5% β-mercaptoethanol, 0.02% bromophenol blue, 30% glycerol, 10% SDS and 250 mM Tris–Cl pH 6.8) for 10 min prior to SDS-PAGE separation (Bio-Rad, 10% acrylamide, 150 V) for subsequent analysis by either Coomassie staining or blotting on nitrocellulose; for prior de-N-glycosylation, samples were pre-treated with PNGase F. For Western blot analysis, the nitrocellulose blot was blocked in 3% bovine serum albumin in 0.5% in PBST or 0.5% BSA in TBST (blocking buffer) at 4 °C. Afterward, the membrane was incubated with either 10 µg/ml AAL or ConA (biotinylated; Vector Laboratories), Dectin-2, DC-SIGN, or hMGL (Fc-conjugated; R&D Systems). After washing with PBST (TBST), blots were incubated with either streptavidin AF790 (Jackson ImmunoResearch) or anti-human IgG Alexa Fluor 690 (Jackson ImmunoResearch); after extensive washing in PBST, the dried membrane was scanned with an imaging system (GE Healthcare Amersham Typhoon).

Proteomic Analysis

Selected bands were cut from the SDS-PAGE gel (~15 µg/lane). After washing and destaining, proteins were fixed in the gel and reduced with DTT and alkylated with iodoacetamide. In-gel digestion was performed with trypsin (Trypsin Gold, Mass Spectrometry Grade, Promega; Trypsin-ultra, MS grade, New England Biolabs) with a final trypsin concentration of 20 ng/µl in 50 mM aqueous ammonium bicarbonate and 5 mM $CaCl_2$ either for 8 h or overnight at 37 °C. Afterward, peptides were extracted thrice with 50 µl of 5% TFA in 50% aqueous acetonitrile supported by ultrasonication for 10 min. Extracted peptides were dried down in a vacuum concentrator (Eppendorf) and resuspended in 0.1% TFA for LC-MS/MS analysis. Peptides were separated on a nano-HPLC Ultimate 3000 RSLC system (Dionex). Sample pre-concentration and desalting was accomplished with a 5 µm Acclaim PepMap µ-Precolumn (300 µm inner diameter, 5 µm, 100 Å; Dionex). For sample loading and desalting, 2% acetonitrile in ultra-pure H_2O with 0.05% TFA was used as a mobile phase with a flow rate of 5 µl/min. Separation of peptides was performed on a 25 cm Acclaim PepMap C18 column (75 µm inner diameter, 2 µm, 100 Å) with a flow rate of 300 nl/min. The gradient started with 4% B (80% acetonitrile with 0.08% formic acid) for 7 min, increased to 31% in 30 min, and to 44% in additional 5 min. It was followed by a washing step with 95% B. The mobile phase A was ultra-pure H_2O with 0.1% formic acid. For mass spectrometric analysis, the LC was directly coupled to a high-resolution Q Exactive HF Orbitrap mass spectrometer. MS full scans were performed in the ultrahigh-field Orbitrap mass analyzer in ranges *m/z* 350 – 2000 with a resolution of 60,000, the maximum injection time was 50 ms, and the automatic gain control was set to 3×10^6. The top 10 intense ions were subjected to Orbitrap for further fragmentation *via* high energy collision dissociation activation over a mass range between *m/z* 200 and 2000 at a resolution of 15,000 with the intensity threshold at 4×10^3. Ions with charge state +1, +7, +8, and >+8 were excluded. Normalized collision energy was set at 28. For each scan, the automatic gain control was set at 5×10^4 and the maximum injection time was 50 ms. Dynamic exclusion of precursor ion masses over a time window of 30s was used to suppress repeated peak fragmentation.

Protein Glycan Analysis

For protein-specific glycan analysis, the glycopeptides were heat treated to inactivate the trypsin and the sample was subject to treatment with PNGase F (250U, Sigma-Aldrich; pH 8) followed by PNGase A (New England Biolabs; pH 6). The released N-glycans were purified using initially a cation exchange material (Dowex AG50 H^+ form, Bio-Rad) and the glycans were eluted using 2% acetic acid, whereas the residual peptides were eluted using 0.1 M ammonium acetate, pH 6.0. N-glycans were purified using a column packed respectively nonporous graphitized carbon/Lichroprep C18. The column was pre-washed with 100% acetonitrile and equilibrated with water. The N-glycans were then eluted with 40% acetonitrile containing 0.1% TFA. After drying, the glycans were fluorescently labeled by reductive amination using 2-aminopyridine and then analyzed with MALDI-TOF MS and RP-HPLC as described above.

Phylogenetic Analyses

Defined *N*-acetylglucosaminyltransferase protein sequences from *Homo sapiens* as annotated by NCBI (MGAT1, P26572; MGAT2, Q10469; MGAT3, Q09327; MGAT4A, Q9UM21; MGAT4B, Q9UQ53; MGAT4C, Q9UBM8; MGAT5A, Q09328) and *C. elegans* as annotated in Wormbase (MGAT1, GLY-12, GLY-13 and GLY-14; MGAT2, GLY-20; MGAT5, GLY-2) were used as a query to search homologous sequences of the whole nematode proteome (NCBI 11.01.2023) using hidden Markov Models algorithm from phmmer (http://hmmer.org/). All found sequences were used to built an alignment with MAFFT (34) prior to building the final approximately-maximum-likelihood phylogenetic tree with the FastTree tool (35). The resulting phylogeny tree was visualized with iTOL (36).

RESULTS

A Natural T. suis *N-Glycan Array*

As a means for understanding glycan-dependent immunomodulatory effects of *T. suis* excretory-secretory products, we undertook the construction of a natural N-glycan array. Enzymatically released glycans were nonreductively labeled with FMAPA, a fluorescent methoxyamino-based linker developed in the Cummings' laboratory and then separated by HPLC and analyzed by MALDI-TOF MS/MS (Fig. 1 and supplemental Figs. S1–S4); prior to printing, the glycans are treated with piperidine, which removes the fluorescent Fmoc moiety and reveals a free amino group which can be printed onto NHS-modified glass slides (29). In total 27 *T. suis* glycan fractions, five control compounds, and a 'buffer only' set of spots were printed in each field.

Standard Lectins

An initial validation of the array was performed using a number of widely used lectins binding a range of glycan motifs

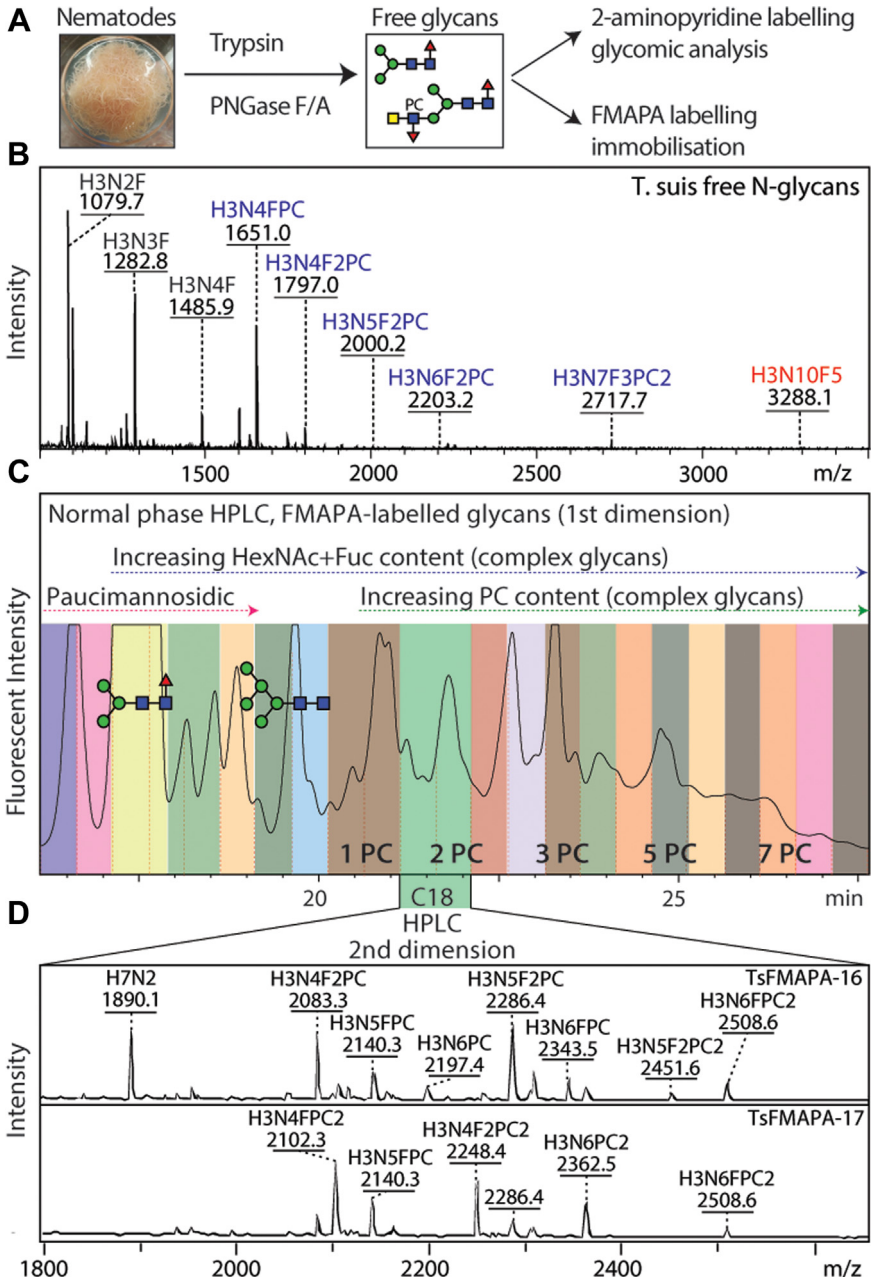


FIG. 1. **Glycomic analysis of *Trichuris suis* N-glycans.** *A* and *B*, N-glycans were released from tryptic peptides and analyzed by MALDI-TOF MS; 90% were labeled with FMAPA for preparation of glycan arrays, the remaining 10% with 2-aminopyridine for in-depth glycomic analysis (see Figs. 3–5 and supplemental Figs. S7–S21). *C*, the FMAPA-labeled glycans (1600 nmol) were fractionated by semipreparative normal phase HPLC (Luna NH2), annotations of a pauci- and an oligo-mannosidic structure are on the basis of MALDI-TOF MS data; the elution positions for complex glycans with increasing numbers of phosphorylcholine residues are also indicated as PC1 – PC7 (supplemental Fig. S1). Selected pools were subject to a second dimension on a reversed-phase column. *D*, the resulting 27 pools, each enriched in a subset of glycans (see mass spectra for examples indicating that each printed fraction contained a number of structures with phosphorylcholine-modified glycans detected as $[M + H]^+$ and other glycans as $[M+Na]^+$), were treated with piperidine prior to array printing (see Fig. 2 and supplemental Fig. S5 for results). For further mass spectra of FMAPA-labeled glycans, refer to supplemental Figs S2–S4.

(Fig. 2*A*; see also supplemental Fig. S5*A* for data). A majority of *T. suis* glycan fractions were bound by the 'fucose-specific' AAL and the 'mannose- and simple complex-specific' ConA (most fractions 1–20); many of the fractions recognized by the latter were also bound by the 'mannose-specific' *G. nivalis* lectin (fractions 1–11, except 10). On the other hand, only a few fractions, concluded to contain glycans with unsubstituted LacdiNAc motifs (B fragments of *m/z* 407, *i.e.*, $HexNAc_2$), were recognized by wheat germ or *W. floribunda agglutinins* (WGA and WFA; *e.g.*, 9 and 14–16). Binding by LTL

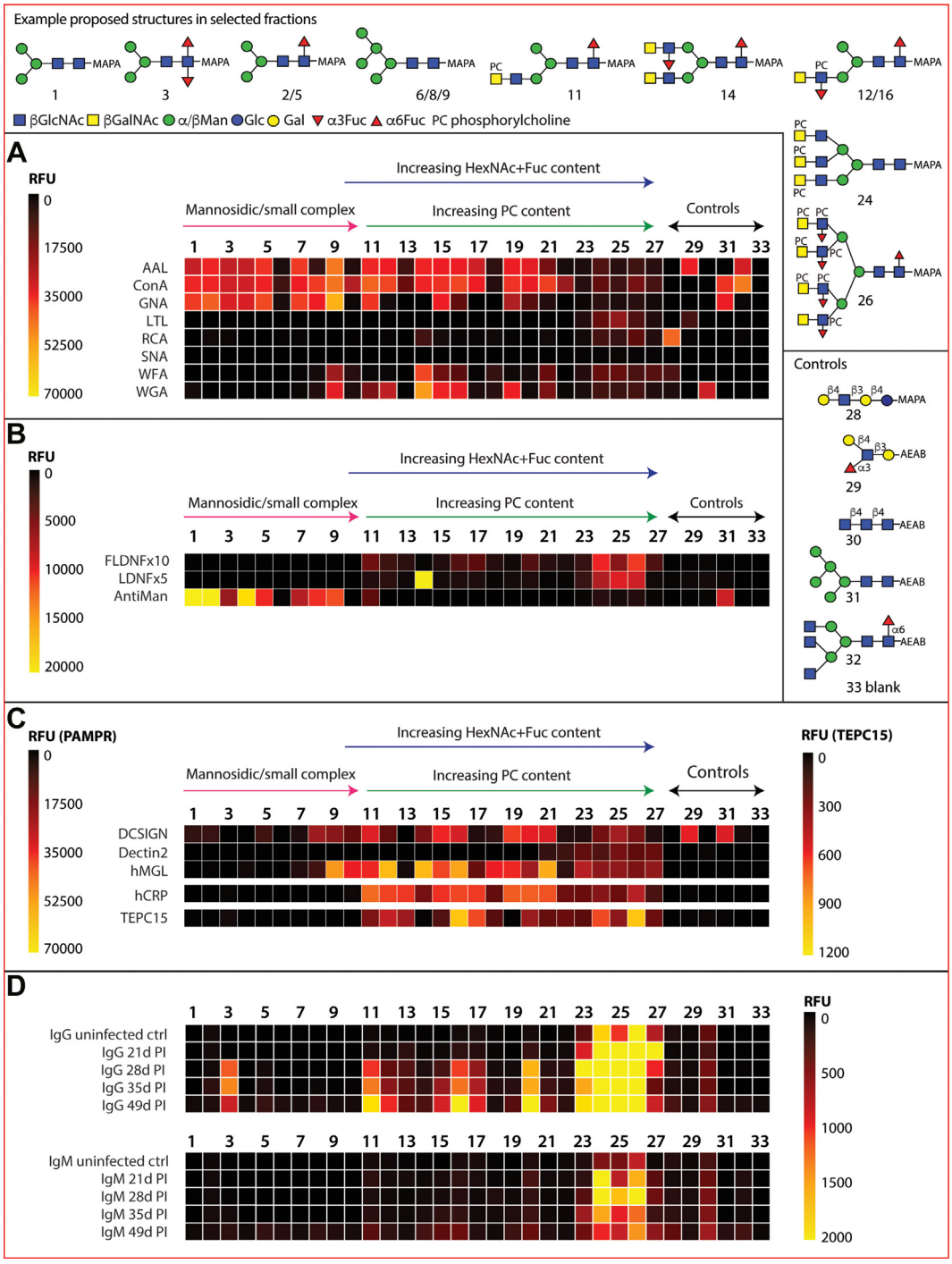


FIG. 2. **Array analysis of *Trichuris suis* N-glycans.** Twenty-seven HPLC fractions of F-MAPA-labeled natural N-glycans derived from *T. suis* (1–27) in addition to five controls (28–32) and a buffer control (33) were printed and incubated with a variety of (*A*) commercial lectins, (*B*) anti-glycan antibodies, (*C*) pathogen-associated molecular pattern receptors, and (*D*) sera from infected pigs. The data are shown as heatmaps representing the relative fluorescence units (RFUs) following the key in each panel; for using a comparable scale for the anti-glycan antibodies, the values for anti-FLDNF (F2D2) and anti-LDNF (L6B8) are multiplied by 10- or 5-times. Example proposed structures for the major components in selected fractions and the pure control glycans are shown according to the Symbol Nomenclature for Glycans (refer also to supplemental Figs. S1–S4 for relevant HPLC and MS data and supplemental Fig. S5 for bar chart representations).

is most obvious for the largest glycans displaying fucosylated antennae (fractions 23–26), whereas recognition by *R. communis* (RCA) or *S. nigra* (SNA) lectins to *T. suis* glycans was relatively weak, which correlates with the glycomic analyses (see below) indicating a lack of antennal galactose or sialic acid.

Anti-Glycan Antibodies

Three monoclonal antibodies originally raised against schistosomal antigens, with reported binding to mono/difucosylated LacdiNAc or paucimannosidic structures, were used to probe the array (Fig. 2*B* and supplemental Fig. S5*B*). The LDNF antibody bound especially to fraction 14, which contains glycans with probable fucosylated LacdiNAc motifs (B fragments of *m/z* 553, *i.e.*, $HexNAc_2Fuc_1$); a lower degree of binding was also observed for the LDNF and FLDNF antibodies to higher molecular weight fractions 24 to 26 containing glycans with putative LacdiNAc motifs decorated with fucose and phosphorylcholine, as judged by B fragments of *m/z* 515 and 718 ($HexNAc_{1-2}Fuc_1PC_1$). The anti-mannose monoclonal had a specificity rather similar to that of GNA, binding best to fractions containing $Man_{3-5}GlcNAc_2Fuc_{0-1}$ as well as the $Man_5GlcNAc_2$ control.

Innate Immune Lectins

Three human lectins, including two known to bind *T. suis* soluble products (27), were used to probe the array (Fig. 2*C* and supplemental Fig. S5*B*). DC-SIGN appeared to bind especially fractions containing oligomannosidic glycans ($Man_{5-9}GlcNAc_2$), overlapping with those bound by ConA. Those fractions recognized by macrophage galactose lectin (MGL) partly overlapped with those of WGA and CRP, suggesting that LacdiNAc-modified glycans (with or without phosphorylcholine, defined on the basis of *m/z* 407 and 572 B fragments) were ligands. Dectin-2 displayed relatively low binding rather to the fractions 23 to 27 containing the largest N-glycans.

Phosphorylcholine-Binding Proteins

CRP and the TEPC-15 IgA monoclonal are well established as being specific for phosphorylcholine, a typical modification of nematode glycans. Here, this specificity is verified (Fig. 2*C* and supplemental Fig. S5*B*), as fractions rich in glycans with *m/z* 369 ($HexNAc_1PC_1$) and related B fragments were recognized, but there are subtle differences. CRP bound a range of fractions with glycans predicted to contain one or two phosphorylcholine residues but relatively less well to those with multiple phosphorylcholine-modified antennae. TEPC-15 binding to various fractions overlapped with that of CRP but was lower in terms of absolute fluorescence intensity.

Pig Infection Sera

IgG and IgM from pigs infected with *T. suis* bound especially to glycans present in a limited number of fractions (Fig. 2*D*), especially those with the highest molecular weight glycans (2500–5000 Da) carrying multiple phosphorylcholine residues as well as to fraction 3 containing a core difucosylated N-glycan. Thereby, the trend is that the antibodies, only partly present prior to infection, bind to those glycan fractions with lower binding to DC-SIGN, MGL, and CRP but high recognition by the anti-FLDNF, anti-LDNF, and anti-PC (TEPC-15) monoclonal antibodies. IgG and IgM binding is highest for days 21 to 28 after infection (supplemental Fig. S5*C*).

Analysis of an Extended Range of T. suis *N-Glycans*

MALDI-TOF MS of the F-MAPA-labeled N-glycans indicated the presence of highly complex structures, but their separation was suboptimal despite two rounds of HPLC; this meant their fragmentation was partly ambiguous due to co-eluting structures. Thus, we undertook a more detailed glycomic analysis of *T. suis*. Previously, we analyzed its N-glycome using one adult individual and observed, in addition to a plentiful paucimannosidic $Man_3GlcNAc_2Fuc_1$ structure, a number of minor glycans carrying one or two antennae modified with LacdiNAc, fucose, and/or phosphorylcholine residues (14). In the present study, we first employed MALDI-TOF MS of permethylated N-glycans; this did indicate the occurrence of larger structures with multiple fucosylated LacdiNAc motifs (supplemental Fig. S6), but phosphorylcholine-modified glycans cannot be analyzed by this method as they are lost during derivatization. Therefore, we labeled the glycans with 2-aminopyridine and performed off-line 2D-HPLC-MALDI-TOF MS (first hydrophilic interaction, then reversed-phase followed by MALDI-TOF MS; Fig. 3 and supplemental Fig. S7) complemented by chemical and enzymatic treatments, as in a previous study on *Dirofilaria* (16); the resulting analyses showed an immense glycomic variety, including many structures predicted to contain multiple phosphorylcholine residues. The typical paucimannosidic and oligomannosidic glycans again dominated the glycome (14) with about 60% being $Man_3GlcNAc_2Fuc_1$ as judged by fluorescence; these glycans were analyzed, but the data are only summarized as they have been found in many other species. Compared to our previous study with limited material, we could now detect bi-, tri-, and tetra-antennary glycans with terminal GlcNAc residues (*m/z* 1541, 1744, 1801, and 1947; Fig. 4*D* and supplemental Fig. S7), which eluted unusually late on the RP-amide column as compared to isomers from *Dirofilaria immitis* (16), suggesting that there was a different antennal configuration. These backbones are the basis for larger structures of 3000 Da and beyond with various categories of antennal motif (Fig. 3 and supplemental Fig. S8).

LacdiNAc and Fucosylated LacdiNAc

MS/MS of various glycans yielded B-fragments of *m/z* 407 and 553 ($HexNAc_2Fuc_{0-1}$; Fig. 4, *B* and *C* and supplemental Figs. S9 and S10), which are commonly observed in various

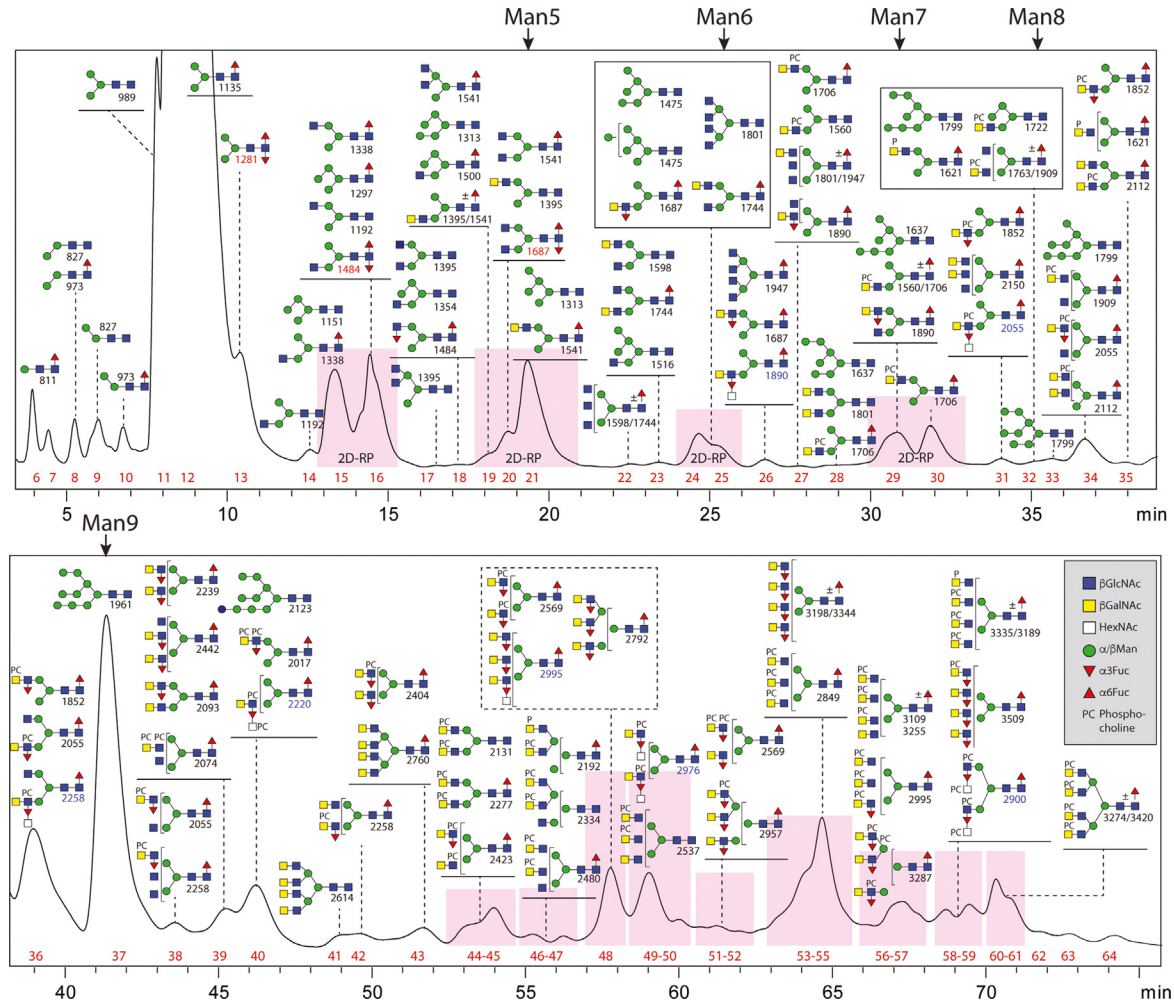


FIG. 3. **Size-fractionation of *Trichuris suis* N-glycans.** HIAX (hydrophilic interaction/anion exchange; AS11 column) HPLC was performed twice on the pool of PNGase F/A-released N-glycans. The order of elution is approximately according to size, but oligomannosidic structures elute rather late as compared to other glycans of similar mass, while HexNAc substitution of antennal fucose reduces retention and phosphorylcholine or phosphate modifications increase retention. Selected fractions were pooled on the basis of MALDI-TOF MS data and rechromatographed on an RP-amide column (see supplemental Figs. S7, S8 and S13 for 2D-HPLC of fractions containing respectively N-glycans of 1100–2000, 2000–3500 Da, and of around 3500 Da). Theoretical *m/z* values, proposed structures and fraction numbers, as well as Glytoucan accessions for glycans up to 2000 Da, are listed in supplemental Tables S1 and S2. Fraction numbers are shown in *red*; masses of glycans with difucosylated cores or HexNAc-substituted antennal fucose are respectively *red* or *blue*. The elution positions for oligomannosidic standards are also indicated (Man5, Man6, Man7, Man8, and Man9).

invertebrates. Such motifs can be based on LacdiNAc (GalNAcβ1,4GlcNAc) or chitobiose (GlcNAcβ1,4GlcNAc). Fortunately, the GalNAc-specific *C. elegans* HEX-4 hexosaminidase can distinguish the two motifs, while *S. plicatus* chitinase does not (16). Based on use of these two exoglycosidases, it could be determined that the $HexNAc_2$ motifs in *T. suis* are solely LacdiNAc (for example digests, see supplemental Figs. S9*A* and S10, *C* and *D*), as compared to the chitobiose or chitotriose motifs found in a variety of other nematodes (16, 37, 38). The fucose residue was sensitive to hydrofluoric acid or almond α1,3/4-fucosidase (39, 40), whereby the antennal fucose residues are α1,3-linked to the subterminal GlcNAc (see examples for *m/z* 1687; supplemental Fig. S9*B*), as previously concluded (14).

Difucosylated Core GlcNAc

We have previously observed that *T. suis* has a simple α1,3/6-difucosylated N-glycan identical to one found in a large range of invertebrates, specifically $Man_3GlcNAc_2Fuc_2$ yielding a characteristic *m/z* 592 Y-fragment upon MS/MS (14) as also observed for one isomer of *m/z* 1687 (Fig. 4*A*); however, also a limited number of larger glycans displayed this modification, including a triantennary glycan of *m/z* 2938, featuring both *m/z* 553 (B-ion) and *m/z* 592 (Y-ion) fragments characteristic of

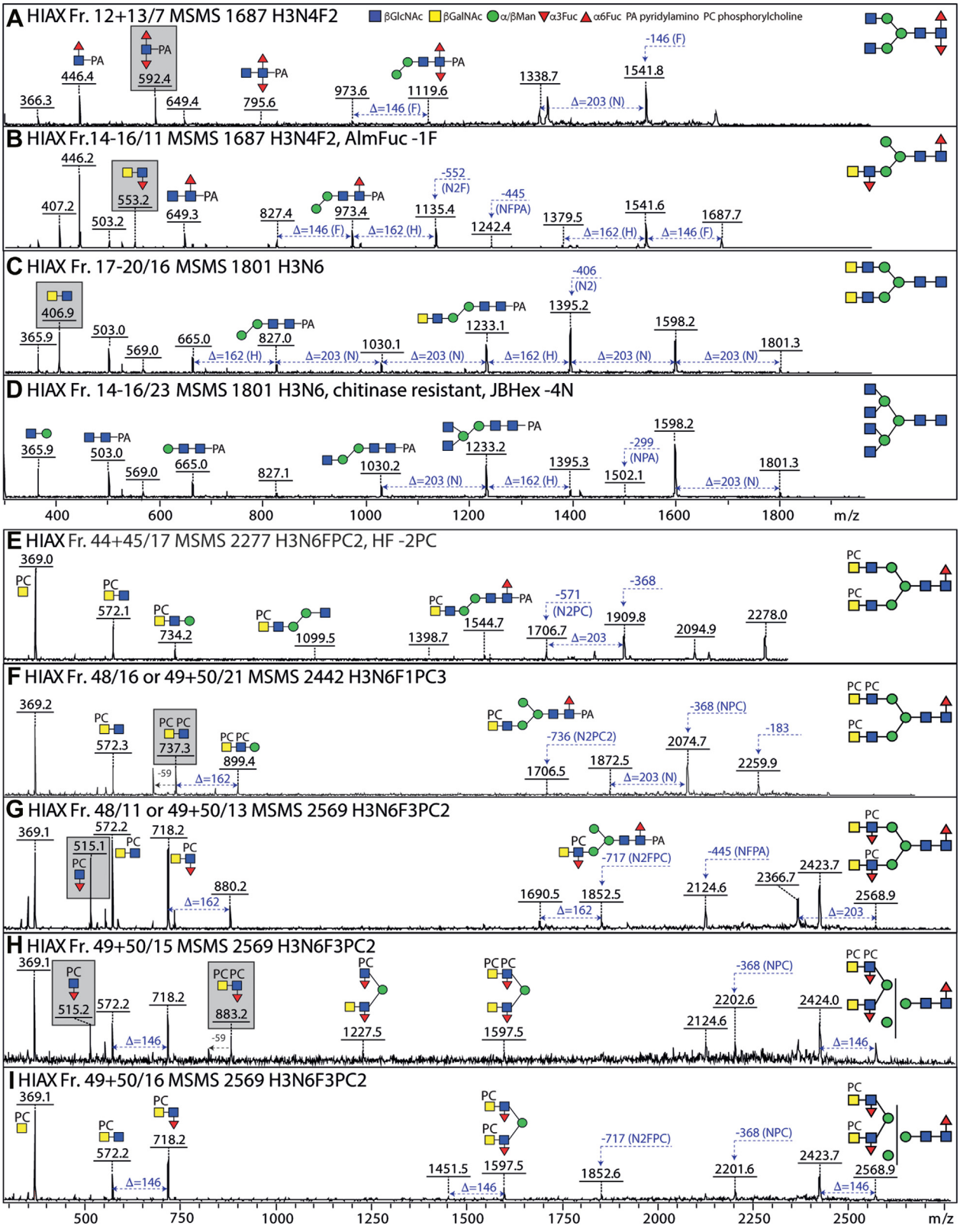


FIG. 4. **Isomeric variations of LacdiNAc-modified, fucosylated, and/or phosphorylcholine-modified N-glycans.** *A–D*, MALDI-TOF MS/MS of selected N-glycans with LacdiNAc and fucose modifications from different 2D fractions (HIAX followed by RP-amide). Shown are isomers of *m/z* 1687 (*A* and *B*), 1801 (*C* and *D*). Core difucosylated structures display the signature *m/z* 592 Y1 ion (*A*), whereas LacdiNAc and antennal

having both fucosylated LacdiNAc and core difucosylation (supplemental Fig. S10*I*).

Phosphorylcholine- and Phosphate-Modified LacdiNAc

Glycans in many nematodes, as well as example cestode and lepidopteran species, are modified with the zwitterion phosphorylcholine 6-linked *via* phosphodiesters to *N*-acetylhexosamine residues (13, 41–43). We have previously reported simple examples from *T. suis* (14), but here we find a very large range of phosphorylcholine-modified structures with the signature *m/z* 369 ($PC_1HexNAc_1$) B-ion fragment. Further B-ions at *m/z* 515, 572 and 718 ($PC_1HexNAc_{1-2}Fuc_{0-1}$; Fig. 4, *E–I* and supplemental Figs. S9*C*, S11-S18) showed the presence of PC-modified LacdiNAc and fucosylated LacdiNAc, whereby two positions for the PC moieties were observed. Often the *m/z* 369 fragment or the loss of 368 Da was very dominant, indicative of a terminal $PC_1HexNAc_1$ unit; however, in the case of an intense *m/z* 572 ion or the loss of 203 Da (*i.e.*, HexNAc), it is concluded that the PC substitutes the subterminal GlcNAc. Similarly, for fucosylated antennae, the occurrence of *m/z* 515 ($PC_1HexNAc_1Fuc_1$) fragments indicated that the subterminal HexNAc was modified with both phosphorylcholine and fucose, while an *m/z* 718 fragment does not specify the position of the phosphorylcholine (compare, *e.g.*, Fig. 4, *G* and *I*). The antennal phosphorylcholine and fucose residues were, in either case, sensitive to hydrofluoric acid treatment, known to cleave phosphodiester bonds (44) as well as α1,3-fucose; the underlying LacdiNAc motifs were digested with either chitinase or *C. elegans* HEX-4 (supplemental Figs. S11 and S15). Also, for the largest glycans of 3000 to 5000 Da, treatment with hydrofluoric acid and chitinase revealed a common late-eluting tetra-antennary backbone structure (supplemental Figs. S13 and S14).

Due to the 2D-HPLC fractionation, many isomeric forms could be detected, including hybrid or pseudohybrid structures with two antennae on the same α-mannose (*e.g.*, with key B3 fragments of *m/z* 1140, 1305, 1451 and 1597; $Hex_{1-}HexNAc_4Fuc_{0-2}PC_{1-2}$) or bi- and tri-antennary structures (Fig. 4, *H* and *I* and supplemental Figs. S12, S16, S17, and S18) as well as the aforementioned tetra-antennary glycans (supplemental Fig. S13). Some structures carrying phosphorylcholine were isobaric with heavily fucosylated ones, for example, glycans of *m/z* 2995 have compositions of either $Hex_3HexNAc_8Fuc_2PC_3$ or $Hex_3HexNAc_9Fuc_4$ as distinguished by retention time and fragmentation (supplemental Fig. S16, *G–J*); for other compositions, up to 10 isomers were detected, for example, $Hex_3HexNAc_6Fuc_2PC_2$ and $Hex_3HexNAc_{7-}Fuc_3PC_2$ (*m/z* 2423 and 2773; supplemental Table S2 and supplemental Figs. S12, *A–F* and S16, *D–F*). In a few cases, *m/z* 737 or 883 fragments were observed, indicative of a LacdiNAc motif, with or without fucose, modified with two phosphorylcholine residues (Fig. 4, *F* and *H*).

In addition, some glycans had an 80 Da modification with relevant fragments or losses in negative or positive mode (supplemental Fig. S17, *A* and *B*), which was sensitive to hydrofluoric acid: thus, it was concluded that these glycans carry a terminal phosphate residue on one of their antennae rather than a phosphorylcholine (supplemental Figs. S11*E*, S13, and S17). This combination has not been previously observed in nematodes.

Substituted Antennal Fucose Residues

In a number of nematodes, such as *Dirofilaria* (16), $HexNAc_3Fuc_1PC_{0-1}$ motifs can be detected based on the relevant *m/z* 756 and 921 Y fragments, which upon hydrofluoric acid treatment are replaced by one at *m/z* 610 ($HexNAc_3$). Puzzlingly, in the case of *T. suis*, no $HexNAc_3$ motifs were observed upon this chemical defucosylation treatment, rather only *m/z* 407 ($HexNAc_2$) motifs, even in the case of large glycans with eight or more HexNAc residues and multiple fucose and phosphorylcholine modifications (supplemental Fig. S14); furthermore, these digests showed that a HexNAc and a fucose were removed at once (Fig. 5, *D–F* and supplemental Figs. S9*A*, S15*E*, S19, *D–F* and S20). We therefore assume that the antennal fucose residues can be substituted by an *N*-acetylhexosamine, which is in turn occasionally modified by phosphorylcholine (*m/z* 921 and 1087 $HexNAc_3Fuc_1PC_{1-2}$ B-ions; Fig. 5, *A–C* and supplemental Figs. S16, S18, S19, and S20). Whereas many examples of the substituted fucose are in the context of LacdiNAc units, some lack the GalNAc of the LacdiNAc; as judged by the absence of *m/z* 572 and 718 B-ions and the presence of *m/z* 515 and 883 ($HexNAc_{1-2}Fuc_1PC_{1-2}$) fragments (supplemental Fig. S19, *G* and *H*), they are concluded to carry PC-modified HexNAc directly linked to the antennal fucose. To date, these specific combinations of modifications have not been observed in other organisms.

fucosylated LacdiNAc are characterized respectively by *m/z* 407 and 553 B2 ions. *E–I*, MALDI-TOF MS/MS of selected N-glycans with phosphorylcholine modifications from different 2D fractions. Shown are one structure each of *m/z* 2277 and 2442 (*E* and *F*) and three isomers of *m/z* 2569 (*G–I*). The pattern of B1 and B2 ions (ratios of *m/z* 369 and 572 or *m/z* 515 and 718) as well as the loss of either 203 or 368 Da is indicative for the position of the phosphorylcholine on LacdiNAc and fucosylated LacdiNAc (either on the terminal GalNAc or subterminal GlcNAc) or whether both zwitterionic residues are on the same antenna (*m/z* 737 and 883). Fractions are named with the HIAX pool number followed, if relevant, by the RP-amide second dimension fraction number (see Fig. 3 or supplemental Figs. S7 and S8), the *m/z*, and the abbreviated composition of the form H3NxFyPCz for $Hex_3HexNAc_{4-6}Fuc_{0-3}PC_{0-3}$. Some fractions were also incubated with exoglycosidases with the noted resistance or sensitivity. *Gray boxes* highlight isomer/isobar-specific B2 ions. For further examples of MS/MS, refer to supplemental Figs. S9–S21.

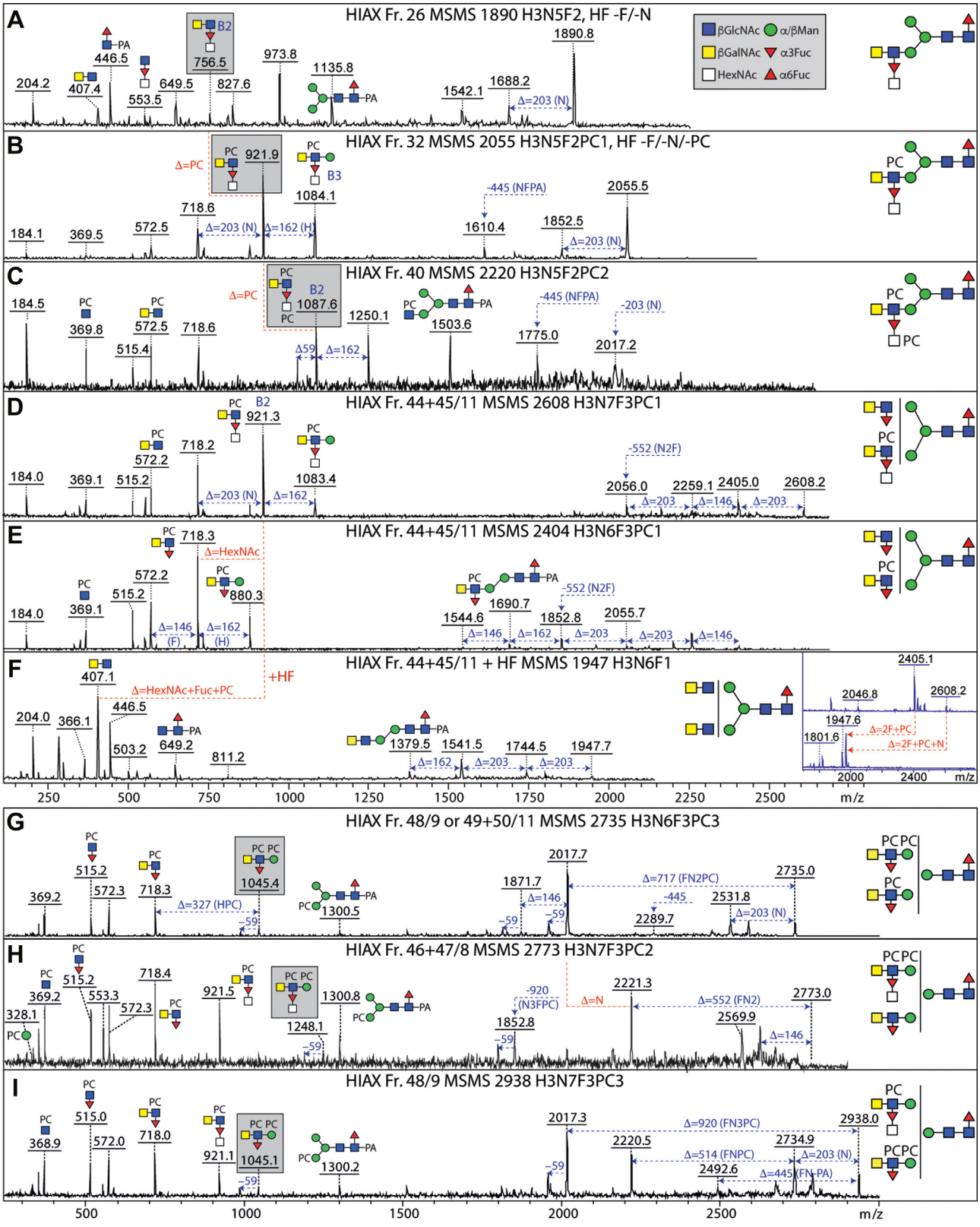


FIG. 5. **N-glycans with antennal *N*-acetylhexosamine-substituted fucose residues and/or phosphorylcholine modifications of mannose.** MALDI-TOF MS/MS of selected N-glycans in different 1D- or 2D-HPLC fractions (see Fig. 3 and supplemental Fig. S8) with fucosylated LacdiNAc modifications carrying further -acetylhexosamine and phosphorylcholine residues. The simplest hybrid structures with

Phosphorylcholine-Modified Core Mannose

Among the various glycans modified with phosphorylcholine, some yielded MS/MS fragments indicative of modification of the core trimannosylchitobiosyl region (*m/z* 1154 and 1300, *i.e.*, 165 Da in addition to 989 or 1135; see Fig. 5, *G–I* and supplemental Fig. S21). Also, mass differences of 327 Da (Hex_1PC_1), rather than the usual 368 Da ($HexNAc_1PC_1$), in addition to low-intensity ions at *m/z* 328 were observed; furthermore, B3-ions at *m/z* 1045 ($HexNAc_2Hex_1Fuc_1PC_2$) suggested that there were phosphorylcholine modifications of core α-mannose residues to which PC-modified fucosylated LacdiNAc units were attached. The *m/z* 1248 fragments ($HexNAc_3Hex_1Fuc_1PC_2$), on the other hand, are consistent with the presence of HexNAc-substituted antennal fucose on some glycans with PC-modified mannose.

O-Glycome

β-Elimination followed by permethylation was employed to examine the O-glycans of *T. suis*. Whereas some structures were reminiscent of the hexose-modified structures from *C. elegans* (*e.g.*, *m/z* 942; $Hex_4HexNAc_1$), others were based on multiple HexNAc residues and could also be fucosylated (Fig. 6). While the *m/z* 994 glycan ($HexNAc_3Fuc_1$) could correspond to a fucosylated LacdiNAc-type structure as proposed for some O-glycans from *Haemonchus contortus* (45), the one of *m/z* 1239 ($HexNAc_4Fuc_1$) may carry a HexNAc-substitution of fucose of the type found in the *T. suis* N-glycome. As the glycans were permethylated, it is not possible to determine whether they were either modified with phosphorylcholine, as recently found for some O-glycans from *C. elegans* (46), or naturally methylated as known from *T. canis* (20).

Phylogenetic Analyses

In order to understand the N-glycan branching patterns, the existing *T. suis* genomic data were searched for homologs of *N*-acetylglucosaminyltransferases involved in eukaryotic N-glycan biosynthesis. Unlike *C. elegans* which possesses three GlcNAc-TI, one GlcNAc-TII, and one GlcNAc-TV homologs, respectively GLY-12,-13,-14,-20 and -2 (47) and *Trichinella spiralis* which is predicted to have one isoform each of GlcNAc-TI, GlcNAc-TII, GlcNAc-TIV, and GlcNAc-TV, *T. suis* is predicted to lack any GlcNAc-TV (supplemental Fig. S22), which may account for the different isomeric form of the tri- and tetra-antennary N-glycans as compared to other nematodes (supplemental Figs. S7 and S13).

Glycoprotein Analyses

Lectin blotting of *T. suis* lysates indicated binding to ConA, AAL, Dectin-2, DC-SIGN, and MGL, which was generally reduced upon pre-incubation with PNGase F (Fig. 7), a finding compatible with the array data. A rather dominant band at 46 kDa was observed and excised from the corresponding Coomassie-stained gels prior to tryptic peptide mapping and glycan analysis. The best match for the proteomic data was to an "uncharacterized" *T. suis* protein related to a secreted Poly-Cysteine and Histidine-Tailed Metalloprotein from *T. spiralis*, which is also N-glycosylated (48). Homologous proteins, also known as p43, have also been identified in *Trichinella pseudospiralis* adult secretome (49), *Trichuris muris* excretory-secretory products (50) and *Trichuris trichuria* egg and female extracts (51). In terms of N-glycomics of the protein band, a typical range of pauci- and oligomannosidic glycans as well as structures carrying LacdiNAc residues with or without fucose and phosphorylcholine were revealed upon MALDI-TOF MS/MS of the HPLC-fractionated N-glycans, thereby representing a subset of the overall N-glycome.

DISCUSSION

Nematodes are amazing in terms of their glycomic diversity and, although it is a relatively well-explored species, *T. suis* is no exception. By combining glycan array and glycomic data, we have sought to close the gap between structure and immunological function of the N-glycans of this parasite. Here, we have significantly expanded the knowledge of the range of N-glycans in *T. suis*, showing the occurrence of tetra-antennary N-glycans with variations on the theme of LacdiNAc and have confidently defined some 200 N-glycan structures, including isomeric and isobaric forms, as compared to the less than 40 previously described (14). This has been accomplished by using far more starting material, but also by 2D-HPLC fractionation prior to MALDI-TOF MS/MS, whereby two labeling methodologies were employed: FMAPA for preparing glycans for printing, but not so suitable for isomeric glycan analysis and PA for the in-depth structural determination, but not compatible with printing. As many further masses over 3500 Da probably correspond to various isomers of

substituted fucose residues (*A–C*) show differences in the numbers of phosphorylcholine residues as indicated by the B2 fragments at *m/z* 756, 921, and 1087; HF treatment of the *m/z* 1890 and 2055 glycans (see supplemental Figs. S9 and S15) resulted in loss of the FucHexNAc unit. The co-eluting biantennary structures (*D–F*) of *m/z* 2404 and 2608 were both digested down to *m/z* 1947 with HF (see inset) accompanied by loss of the B2 ions at *m/z* 718 and 921, indicative of the loss of a FucHexNAc unit from the latter. *G–I*, while the typical PC-modified fucosylated LacdiNAc motif (B2; *m/z* 718), also in substituted form (*m/z* 921), shows the occurrence of antennal phosphorylcholine, putative Y fragments at *m/z* 1154 and 1300 are compatible with a PC modification of a core α-mannose residue. Furthermore, B fragments at *m/z* 1045 or 1248 as well as *m/z* 328 are also indicative of PCMan motifs; the occurrence of two phosphorylcholine residues on juxtaposed residues results in the formation of ions 59 Da smaller than the true B ion (fragment ions at *m/z* 986 and 1189). *Gray boxes* highlight isomer/isobar-specific B2 and B3 ions. For further examples of MS/MS, refer to supplemental Figs. S9–S21, including other isomers of *m/z* 1890 and 2773 in supplemental Figs. S10 and S16.

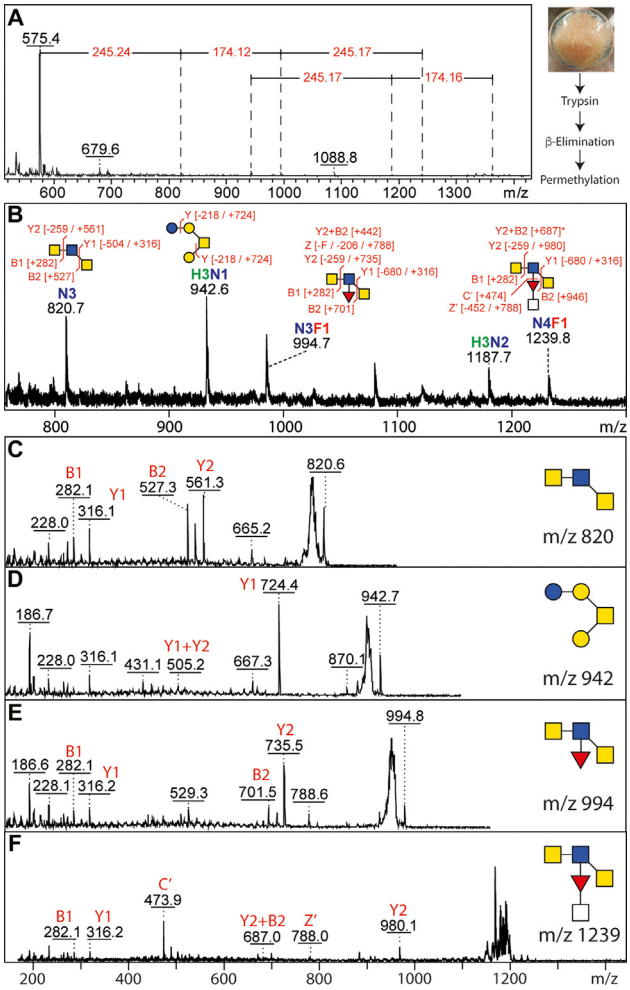


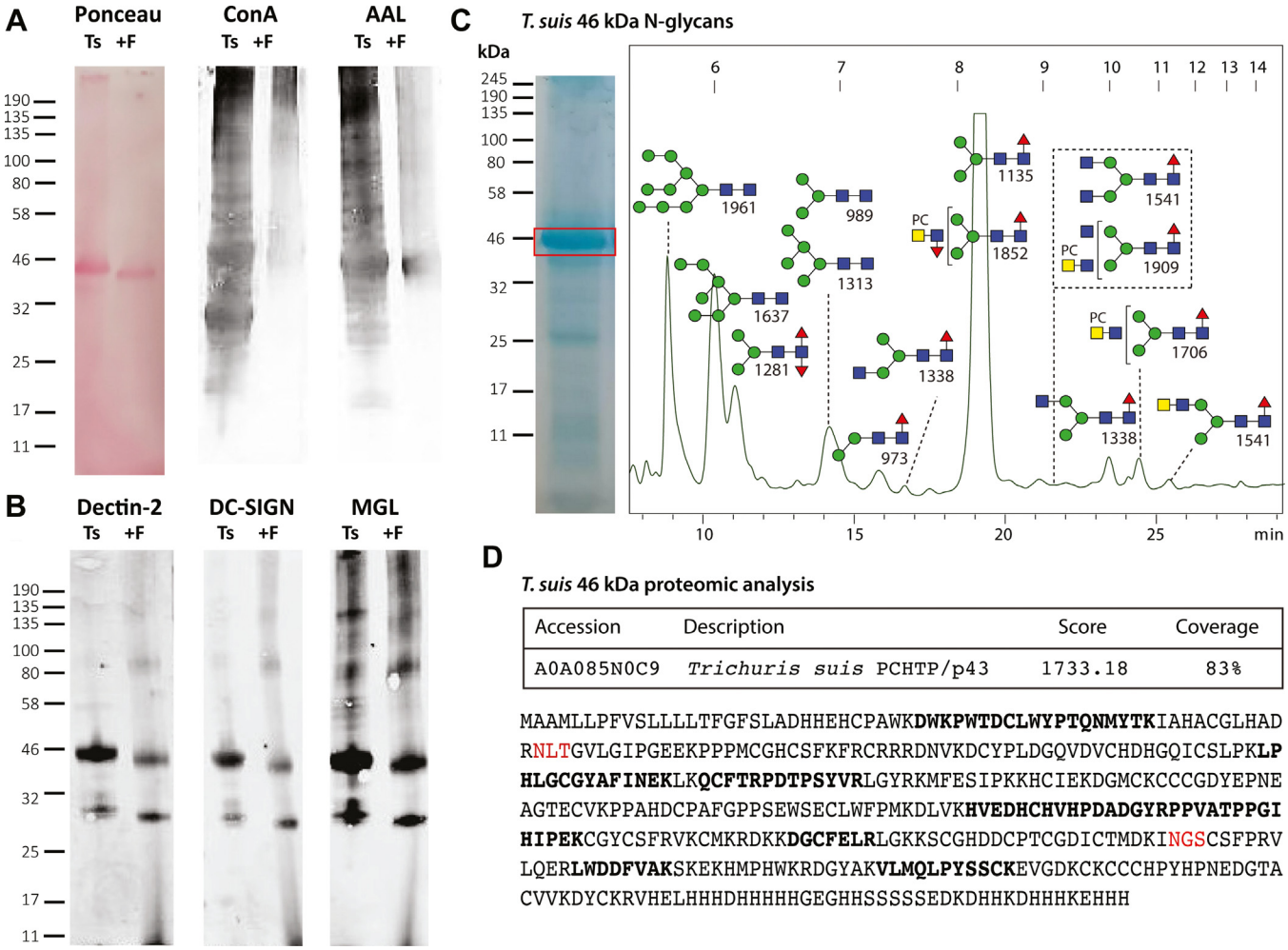
FIG. 6. **MALDI-TOF MS analysis of permethylated O-glycans.** *A*, glycans released by reductive β-elimination were permethylated and analyzed by MALDI-TOF MS, which showed a potential series of up to *m/z* 1362 can be interpreted as being based on HexNAc and Fuc modifications of $HexNAc_2$ and $Hex_3HexNAc_1$ structures. *B–F*, the *m/z* range 800 to 1300 showed four glycans with interpretable MS/MS; while the *m/z* 820, 994, and 1239 all have terminal HexNAc residues (loss of 259 Da), the *m/z* 474 fragment in the latter spectrum is compatible with a HexNAcFuc branch.

highly substituted glycans, but MS/MS in the higher mass range did not result in full structural assignments, the actual number of N-glycan isomers will significantly exceed 200.

On the antennae of the higher molecular weight N-glycans, fucosylated versions of LacdiNAc as well as multiple phosphorylcholine modifications were observed (for a summary of glycan epitopes in *T. suis*, see Fig. 8). As previously found in *T. suis* and *D. immitis* (14), where there was a single modification of a non-fucosylated LacdiNAc, the phosphorylcholine tended to be on the terminal GalNAc rather than on the subterminal GlcNAc, which is in contrast to *C. elegans* (46); in the case of fucosylated LacdiNAc, on the other hand, the zwitterionic moiety was often on the GlcNAc, but $HexNAc_2Fuc_1PC_2$ antennae were found as well as phosphorylcholine modifications of the HexNAc substitution of fucose and of the underlying mannose. The largest detected FMAPA-glycan (supplemental Figs. S1 and S4; *m/z* 5059) would be predicted to contain nine zwitterionic moieties on a $Hex_3HexNAc_{10}Fuc_5$ scaffold, but a higher degree of substitution cannot be ruled out. Certainly, our data indicate multiple positions for phosphorylcholination on *T. suis* tetra-antennary N-glycans (two on each of the maximally four antennae, in addition to one substitution of mannose), whereby the MS/MS fragmentation for the largest glycans is primarily limited to B-fragments. The theme of LacdiNAc, modified by phosphorylcholine or fucose, in the context of tetra-antennary glycans is shared with the related *T. spiralis* (52); however, due to technical limitations in early studies, the degree of phosphorylcholination directly observed in *T. spiralis* was lower than that detected here in *T. suis*. Zwitterionic modification of mannose is rarer, but has been recently reported for one other nematode, *Brugia malayi*, and in filamentous fungi (53, 54).

The one modification in *T. suis* that can be considered 'species-specific' is the modification of the fucose residue of the fucosylated LacdiNAc motif with a further HexNAc, often in combination with a phosphorylcholine modification. Although there are many instances of substituted core fucose on N-glycans from nematodes (45, 46) and molluscs with galactose or disubstitution with hexuronic acid and another monosaccharide (55, 56), this is the only instance of an *N*-acetylhexosamine substitution of fucose in nematodes. On the other hand, unlike *C. elegans*, *Oesophagostomum dentatum*, *H. contortus* or filarial nematodes, there is no evidence for extended chito-oligomer chains (37, 38, 45, 57), but akin to *D. immitis*, the N-glycan core region in *T. suis* is relatively unmodified (primarily α1,6-fucosylation and a trace of α1,3-fucosylation) and lacking the galactosylated fucose or the methylated epitopes found in *O. dentatum* or especially the highly modified cores in *C. elegans* (37, 46). Also, other than some phosphorylated structures, no anionic N-glycans such as the glucuronylated ones found in *D. immitis* or *B. malayi* were detected (16, 53). Due to the phosphate-modified glycans having the same 'backbone' structure as those carrying phosphorylcholine, we assume either that there is an artefact due to the initial sample preparation (reduction and alkylation prior to trypsin digestion) or that a cholinesterase has removed the choline moiety, leaving a phosphate residue; we have not observed such a remodeling in other species, but cholinesterases and phosphorylcholine hydrolases are known from other nematode species (58).

The glycan array is one of a few based on natural N-glycans from a nematode, others being focused on *C. elegans*, *D. immitis* and *B. malayi* (16, 53, 59). Beyond using the typical commercially-available plant and fungal lectins, we probed the arrays with antibodies recognizing glycan modifications and with proteins of the innate immune system. The typical lectins ConA, GNA and AAL recognized a variety of glycans as



Accession	Description	Score	Coverage
A0A085N0C9	*Trichuris suis* PCHTP/p43	1733.18	83%

MAAMLLPFVSLLLLTFGFSLADHHEHCPAWK**DWKPWTDCLWYPTQNMYTK**IAHACGLHAD
RNLTGVLGIPGEEKPPPMCGHCSFKFRCRRRDNVKDCYPLDGQVDVCHDHGQICSLPK**LP**
HLGCGYAFINEKLK**QCFTRPDTPSYVR**LGYRKMFESIPKKHCIEKDGMCKCCCGDYEPNE
AGTECVKPPAHDCPAFGPPSEWSECLWFPMKDLVK**HVEDHCHVHPDADGYRPPVATPPGI**
HIPEKCGYCSFRVKCMKRDKK**DGCFELR**LGKKSCGHDDCPTCGDICTMDKINGSCSFPRV
LQER**LWDDFVAK**SKEKHMPHWKRDGYAK**VLMQLPYSSCK**EVGDKCKCCCHPYHPNEDGTA
CVVKDYCKRVHELHHHDHHHHHGEGHHSSSSSEDKDHHKDHHHKEHHH

FIG. 7. **Western blotting and glycomic analysis of the major *Trichuris suis* glycoprotein.** *A*, *Trichuris suis* extract was subject to SDS-PAGE followed by Western blotting with Concanavalin A and *Aleuria aurantia* lectin before and after PNGase F treatment; Ponceau S staining indicates a shift in the molecular weight of the major protein upon PNGase F digestion. *B*, Western blotting with three innate immune system proteins, Dectin-2, DC-SIGN, and MGL showing partial loss of staining after PNGase F treatment. *C*, coomassie blue staining of SDS-PAGE indicating the excised band of 46 kDa, which was subject to trypsin and PNGase A digestion, prior to labeling of the released N-glycans with 2-aminopyridine and off-line RP-HPLC/MALDI-TOF MS analysis; the MS/MS-verified structures are annotated. *D*, online LC-MS analysis of the tryptic peptides of the 46 kDa band indicate the highest score to a homolog of Poly-Cysteine and Histidine-Tailed Metalloproteins (PCHTP, otherwise known as p43; theoretical mass including signal peptide of 46.7 kDa) found in other Trichinellid species with an overall coverage of 83%; peptides with between 10 and 120 peptide spectrum matches are indicated in *bold* and potential N-glycosylation sites are in *red*.

expected, whereas fewer fractions were bound by WGA and even fewer by WFA; the latter is a trend previously seen for binding to *T. suis* soluble products (27). Considering the abundance of antennal fucose, LTL bound fewer fractions than may be expected, but the high degree of modification of fucosylated LacdiNAc motifs in *T. suis* may interfere with binding. On the other hand, RCA predominantly bound a galactosylated control on the array and recognition by SNA was relatively low as compared to other standard lectins, compatible with a lack of sialic acid (Fig. 2 and supplemental Fig. S5*A*). Overall, the results with these lectins are in keeping with our own previous dot-blot based studies (60) as well as a recent large-scale reanalysis of glycan array data (61).

For the antibodies binding mono- and di-fucosylated LacdiNAc (LDNF and FLDNF) motifs, it was perhaps no surprise that the anti-LDNF antibody (30) bound a number of fractions containing LacdiNAc-modified glycans, including those with phosphorylcholine-modified forms of this epitope, but especially to fraction 14 containing a $Hex_3HexNAc_6Fuc_2$ structure (supplemental Figs. S3 and S5*B*); the case of anti-FLDNF (31) is puzzling, as we detected no difucosylated LacdiNAc motif, but perhaps the multiple fucosylated antennae can also generate an epitope. Amongst the lower molecular weight N-glycans, the anti-mannose 100-4G11-A antibody (32) bound, as expected, fractions containing $Man_{3–5}GlcNAc_2$ and $Man_{3-}GlcNAc_2Fuc_{1–2}$. The TEPC-15 IgA antibody and the pentraxin

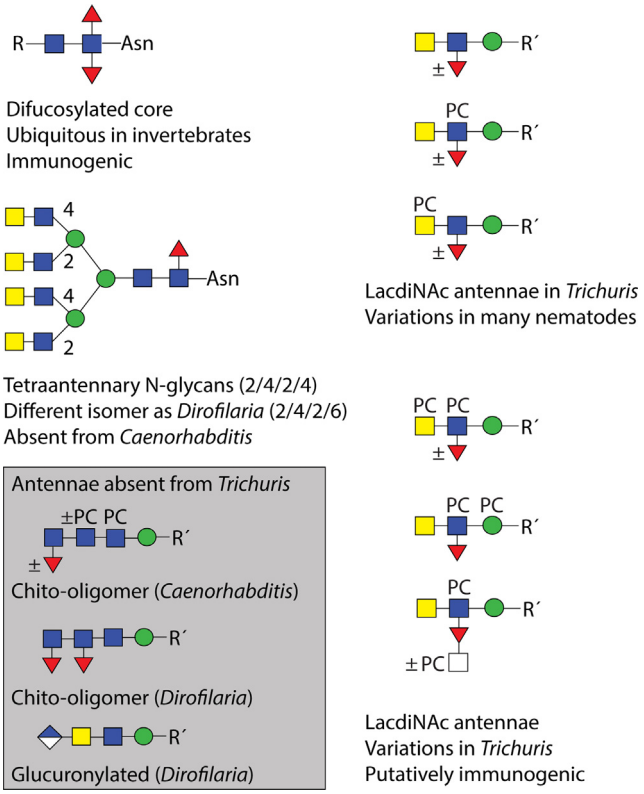


FIG. 8. **Summary of N-glycan motifs in *Trichuris* and selected other nematodes.** Based on the N-glycomic data, the basic elements found or absent from *Trichuris suis* as compared to *Caenorhabditis elegans* and *Dirofilaria immitis* are shown. Glucuronylation, phosphorylcholine-modification of mannose, and fucosylated LacdiNAc have also been reported in *Brugia malayi* or *Trichinella spiralis*. The presence of specific epitopes in other nematodes cannot be excluded and more variations may remain to be detected.

C-reactive protein bound almost all of the fractions judged to contain phosphorylcholine-modified N-glycans in keeping with their known specificity (62).

Of the lectins of the innate immune system examined (Fig. 2), DC-SIGN recognizes a rather wide range of oligomannosidic and fucosylated ligands (63–65); in the present study, its binding correlated generally with the occurrence of $Man_{5-9}GlcNAc_2$ in the relevant fractions. This does not rule out binding to fucosylated LacdiNAc motifs as there was also recognition of some of the largest molecular weight glycans, but it has been shown that glycans with a terminal GlcNAc on the α1,3-mannose with or without a LacdiNAc antenna on the α1,6-mannose are better recognized than those carrying LacdiNAc on the α1,3-mannose (65). Dectin-2 binding was lower on the array relative to other innate immune system lectins, despite reports that it binds larger oligomannosidic structures (66), although it bound well to the major 46 kDa glycoprotein (Fig. 7) and in absolute terms recognized the larger molecular weight glycans (supplemental Fig. S5*B*). MGL, which is known to interact with LacdiNAc (67), bound the majority of fractions, regardless of the presence of phosphorylcholine-modifications of the putative LacdiNAc-containing ligands; the Western blotting results with MGL showing recognition of a 46 kDa band also correlated with the 'band-specific' glycomic data showing presence of LacdiNAc motifs (Fig. 7). In terms of immune modulation by *T. suis* soluble products, it was concluded that their DC-SIGN-mediated effect on dendritic cells is indeed rather *via* the oligomannosidic glycans, whereas MGL may have a role in interactions of these soluble products with other cell types (27). Our data show an obvious increase in antibodies recognizing phosphorylcholine- and fucose-modified N-glycans of 2500 to 5000 Da during *T. suis* infection as compared to uninfected animals (Fig. 2 and supplemental Fig. S5*C*); binding to PC-modified glycans is in line with a number of studies on identified cestode, nematode and trematode antigens (68–72), although in some cases phosphorylcholine appears only to have a minor effect on IgG reactivity (73). Additionally, we observed binding to the widespread invertebrate core difucosylated motif, which is a known epitope for IgE in *Haemonchus*-infected sheep (74) and for anti-horseradish peroxidase (75). In terms of glycan array data for other helminths, the best studied example is the trematode *Schistosoma mansoni*, with which *T. suis* shares the LDNF and core α1,3-fucose motifs. Data from shotgun and defined arrays also show binding of IgG of *Schistosoma*-infected animals to various fucosylated antennal motifs (31, 76, 77), while an immune response to core α1,3-fucose (78) may be restricted to the IgG4 subclass (77).

Our data on the O-glycans is admittedly not as detailed as that for the N-glycans, but are reported here as there is a relative lack of knowledge about nematode O-glycomes. Considering the N-glycan motifs of *T. suis*, it is of interest that fucosylated LacdiNAc may also be a structural motif in its mucin-type oligosaccharides (Fig. 6). As the N-glycans of *T. suis* are not methylated, it is not expected that the permethylation procedure hides the presence of natural methyl groups as known on O-glycans of *Toxocara* (20), a distantly related parasite; however, due to loss of zwitterionic moieties upon permethylation, our data do not answer whether *T. suis* O-glycans carry phosphorylcholine. Other approaches, including non-reductive release, are required to analyze these structures in a native form, also in terms of generation of an O-glycan array.

The major protein band was shown to contain a homolog of poly-cysteine and histidine-tailed metalloproteins (Fig. 7), otherwise known as p43. This protein binds glycosaminoglycans and interleukin-13 (50), thus it represents a potential immunomodulator; in its native form, p43 is also a potent inducer of protective immunity, whereas the insect-derived recombinant form is not (28), but is a cryptic antigen in terms of natural infection (50). As noted above, the band-specific glycomic analysis indicates that p43 may well carry LacdiNAc-containing motifs with and without phosphorylcholine (Fig. 7*C*), which is indeed a type of glycan modification found to a low level in insect cells (13); thus,

reengineering of these cells to increase the proportion of nematode-like glycans would in theory generate p43 in a more native form and so increase the protective effect.

Overall, the N-glycome of *T. suis* encompasses both simple and complex structures, with many modifications being variations of the LacdiNAc unit found in many invertebrates. Multiple antennal fucose (maximally one per antennae or four per glycan) and phosphorylcholine residues (at least up to seven per glycan) in addition to core fucosylation are possible; the combinations found are, to date, novel in *T. suis*, even if most of the modification types are represented in other species (Fig. 8). This leads to the question as to which glycan-modifying enzymes are encoded by its genome. In *Trichuris* spp., there are obvious homologs to three branching *N*-acetylglucosaminyltransferases (supplemental Fig. S22) indicating a different branching pattern as compared to *D. immitis*, as well as possibly 10 α1,3-fucosyltransferases and one putative BRE-4-type LacdiNAc-forming enzyme; however, no nematode phosphorylcholinyltransferase has been identified to date and a fucose-modifying HexNAc-transferase would be novel. Such enzymes are of especial interest as they are responsible for biosynthesis of the glycans which are recognized by pentraxins or antibodies in the sera of infected animals, but which are also associated with immunomodulation; on the other hand, chemical syntheses of phosphodiester-modified glycans are far from routine. Thus, the availability of larger quantities of defined PC-glycan conjugates is limited until a chemoenzymatic approach becomes realizable. The combination of glycomics and glycan arrays presented here is a further step in the definition of complex nematode N-glycans and their interactions with host immune systems, while reinforcing the need for new tools to determine the role of these key post-translational modifications in the evolutionary success of this large group of often parasitic organisms.

DATA AVAILABILITY

The glycomics data is available *via* Glycopost (79) (GPST000360): https://glycopost.glycosmos.org/entry/GPST000360.

The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium *via* the PRIDE partner repository (15) with the dataset identifier PXD045512 and 10.6019/PXD045512: http://central.proteomexchange.org/cgi/GetDataset?ID=PXD045512.

Supplemental Data—This article contains supplemental data (13, 16, 38, 41, 42, 46, 52, 80–82).

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Abbreviations—The abbreviations used are: AAL, Aleuria aurantia lectin; ConA, Concanavalin A; CRP, C-reactive protein; LTL, Lotus tetragonolobus lectin; MGL, macrophage galactose lectin; SPE, solid-phase extraction.

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